THINGS TO KNOW:

1. **Lab report must contain following sections : (order must be maintained)**
   a) **Title /Question**
   b) **Theory: The brief overview of the concept /formula/techniques/technology used in the program**
   c) **Code: The lines of code**
   d) **Output: Screenshot of the output**
2. **Output screen should be captured (use: snipping tool), printed and attached in the report [should be readable]. Other contents must be handwritten.**
3. **Every Source code must include the printing statements to print following information after your main output:**

   Lab No.:

   Name:

   Roll No./Section:

4. **Contents should be written on single side of A4 sized paper.**
5. **The works assigned this week must be submitted next week.**
6. **Cover page and index page should be attached in the report appropriately.**

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<u>Index Page Format (can be printed)</u>

List of Lab works(Numerical Methods)

Lab No.	**Title /Question**	**Submission Date**	**Signature**
1	Solution of nonlinear equations		
1(a)	Bisection Method	2078/12/2	
1(b)	False Position Method	2078/12/2	

Sample output:

```
D:\HariSharma\OOP Lab\sampleProgram.exe
 This is sample output!

 ------------------------------
Lab No.:1(a)/Name: Hari Sharma/Roll No.: 102/Section:C
------------------------------
Process exited after 0.3968 seconds with return value 0
Press any key to continue . . .
```

Language to use: C or C++
Lab Works (Numerical Methods) [Part-1]

1. **Solutions of nonlinear equations.**

   [Output should show the results of each iteration in tabular format.]

   a) WAP in C to find the roots of following equation using **Bisection Method**. (Take the values of intervals (brackets or initial guesses) and error tolerance from user)

   $$x^2 - 4\cos(x) = 0$$

   b) WAP in C to find the roots of following equation using **False Position Method**. (Take the values of intervals (brackets or initial guesses) and error tolerance from user)

   $$x^3 - 3x + 1 = 0$$

   c) WAP in C to find the roots of following equation using **Secant Method**. (Take the values of intervals (brackets or initial guesses) and error tolerance from user)

   $$e^{-x} - 3x = 0$$

   d) WAP in C to find the roots of following equation using **Newton Raphson Method**. (Take the values of initial guess and error tolerance from user)

   $$\cos(x) - x\,e^{x} = 0$$

e) WAP in C to find the roots of following equation using **Fixed Point Iteration Method**. (Take the values of initial guess and error tolerance from user)

$$1 + 1/2 \sin(x) - x = 0$$

2. WAP to implement **Horner's method to evaluate the polynomial** and test for some polynomials.
3. **Implementation of some interpolation techniques.**
   a) WAP to implement **Lagrange Interpolation** and test for some data.
   b) WAP to implement **Newton's Divided Difference Interpolation** and test for some data.
   c) WAP to implement **Newton's Forward Difference Interpolation** and test for some data.
   d) WAP to implement **Newton's Backward Difference Interpolation** and test for some data.
   e) Write a C program for **Cubic Spline Interpolation** and test for some data.
4. **Implementing least square methods for curve fitting (regression analysis).**
   a) WAP to implement **linear least square method (linear regression)** and test for some data.
   b) WAP to implement **nonlinear least square method using exponential model (nonlinear regression with exponential model)** and test for some data.
   c) WAP to implement **nonlinear least square method using polynomial model (nonlinear regression with polynomial model)** and test for some data.

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List of Lab works

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1(b)	False Position Method	2078/12/2	

Language to use: C or C++

5. WAP to implement numerical differentiation techniques using

   a) Forward Difference Formula and Backward Difference Formula (for 1st Derivative)

   b) Central Difference Formula (for 1st derivative and 2nd Derivative)
6. WAP to find the **derivative of a tabulated function** using
   a. Newton's Divided Difference Formula
   b. Newton's Forward Difference Formula
   c. Newton's Backward Difference Formula
7. WAP to compute the **numerical integration** using
   a. Trapezoidal Rule
   b. Composite Trapezoidal Rule
   c. Simpson's 1/3 Rule
   d. Composite Simpson's 1/3 Rule
   e. Simpson's 3/8 Rule
   f. Composite Simpson's 3/8 Rule
   g. Gauss Legendre two point and three-point formula
   h. Romberg integration formula
8. WAP to implement
   a. Romberg Integration Formula
   b. Numerical Double Integration using Trapezoidal Rule.
   (Use any appropriate integral and limits)
9. WAP to solve **ordinary differential equations** using
   a. Taylor's series Method
   b. Picard's Method
   c. Euler's Method
   d. Heun's Method
   e. Runge – Kutta Method

   (Use any appropriate equations)

10. Solutions of system of linear equations using some direct methods

a) WAP to implement **Naive Gauss Elimination Method** to solve the system of linear equations and test it.

b) WAP to implement **Gauss Elimination with Pivoting Method** to solve the system of linear equations and test it.

c) WAP to implement **Gauss - Jordan Method** to solve the system of linear equations and test it.

11. WAP to **inverse a given matrix** using **Gauss - Jordan Method.**

12. WAP to implement following for matrix factorization methods

a. Doolittle's methods

b. Cholesky Decomposition.

13. WAP to **solve system of linear equations** using

a. Matrix inversion method.

b. Doolittle LU Decomposition Method

c. Cholesky Decomposition Method

14. Solutions of system of linear equations using some iterative methods

a) WAP to implement **Jacobi Iteration Method** to solve the system of linear equations and test it.

b) WAP to implement **Gauss-Seidel Method** to solve the system of linear equations and test it.

15. WAP to find the **eigenvalue** and **eigenvector** of a matrix using **Power Method.**

16. WAP for solving boundary value problem using shooting method.

17. WAP to solve partial differential equations using

a. Laplace's equation

b. Poison's equation

(Use any appropriate equations)